Moving Wave and Wave Equation

SYNOPSIS

A spring-mass system can undergo oscillatory motion, and Hooke's Law combined with Newton's Second Law enables us to write an equation of motion whose solution is described by the simple harmonic motion, SHM (see below).

BEFORE YOU READ ON — PREDICT

A pulse runs along a stretched string at some speed. Three things you could change: **pull the string tighter**, **swap it for a heavier string of the same length**, **shake it faster**. Which of those change the speed of the pulse? Write down your three answers. The last section derives the speed from scratch and settles all three at once.

PERFORMANCE EXPECTATIONS

1. **Write** a travelling wave as $\psi(x, t) = A \sin(kx - \omega t)$ and say what k and ω measure.
2. **Explain** why any function of $(x - vt)$ describes a shape travelling at speed v without changing form.
3. **Derive** the wave equation from the forces on a small element of a stretched string.
4. **Read** the wave speed off the wave equation, and connect it to $\sqrt{(F_T/\mu)}$.
5. **Relate** k, ω, λ, f and v to one another, and convert freely among them.

Notes 4 and 5 got that equation of motion and solved it:

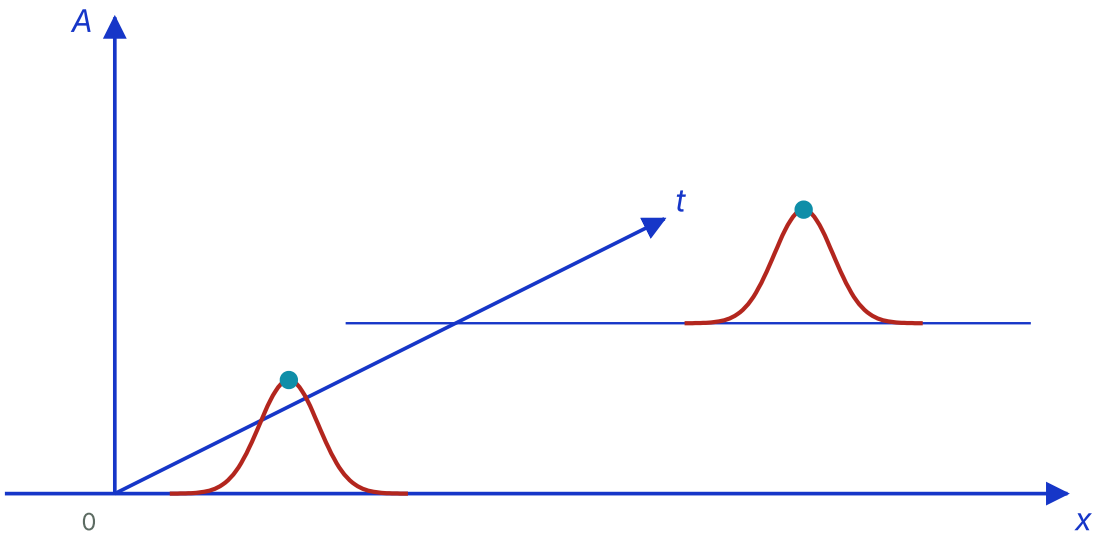
$$d^2x/dt^2 = -(k/m)x \quad \text{with solution} \quad x(t) = A\cos(\omega t + \varphi)$$

where ω is the angular frequency in rad/s and φ is the phase angle.

The SHM describes the amplitude of the back-and-forth oscillation as a function of time only. Since the oscillations occur at a fixed position, say, $x = 0$ on an x-axis (so the amplitude variation is in the y-axis), the amplitude does not vary along the x-axis. The oscillations or the "wave motion" does not "move".

1. Moving wave

Now, let's consider ocean waves or waves on a string. Here, the "disturbances" do move in space or along the string. To see this, imagine taking a snapshot of the string at one time and taking another snapshot at another time. When you compare the two, you will find that the wave is simply shifted in space and the amount of shift in space will depend on the speed at which the wave moved. Mathematically, we can denote this shift as follows:



Join the two peaks and drop each one to the x axis. Mark the distance the pulse has moved and write what that distance is in terms of v and t.

1. Call the shape at $t = 0$ simply $A(x)$. If the whole shape travels a distance vt to the right, what is the height at position x at time t, written in terms of the original function?

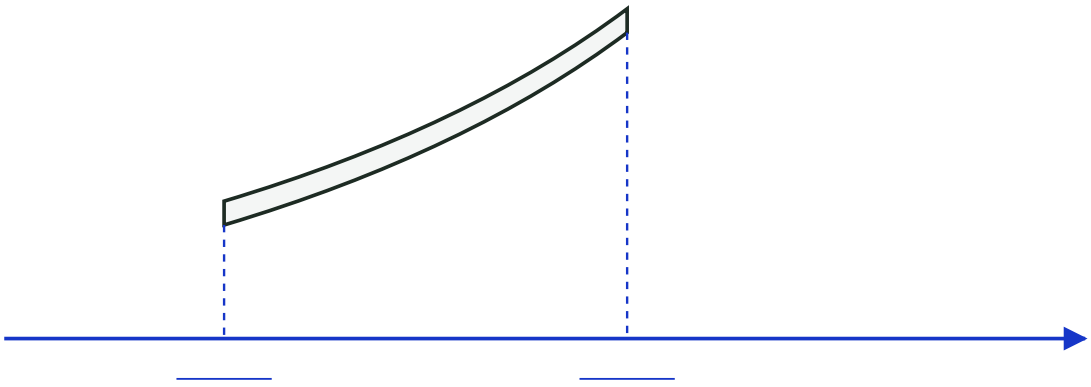
2. What general form must **any** rightward-moving wave therefore have?

3. Check it against a familiar one: show that $A(x, t) = A_0 \cos(kx - \omega t)$ has that form, and read off its speed.

4. Rewrite that wave using the wavelength and the frequency instead of k and ω.

2. Wave equation

Section 1 says what a moving wave *looks like*. It does not say why a string should carry one, or how fast. For that, do to a piece of string exactly what Note 4 did to a block: write down Newton's second law for it. We are deriving an equation of motion for a mechanical wave.



The string has mass μ per unit length. Resolve the tension at each end of the element into a horizontal and a vertical part. Label the two positions, mark which parts cancel, and name what is left.

5. Write the mass of the element and its vertical acceleration, using D for the displacement of the string. Then write ma for the element.
6. Write the net vertical force as the sum of the vertical parts of the two tensions. What happens to the horizontal parts, and why does that matter?
7. Express each vertical part using the **slope** of the string at that end. Mind the sign at the left-hand end.
8. Put the two sides together, divide by $\mu\Delta x$, and let $\Delta x \to 0$. What is the bracket in the limit?
9. Check the units of T_h/μ. What must that combination be?

THE PUNCHLINE

The **wave equation**, with $v^2 = T_h/\mu$ for a string. Two things are worth seeing at once. **The speed came out of the string** — its tension and its mass per length — and nothing about the shaking appears in it. And **ψ is deliberately not called D**: the same equation turns up for sound, for light, for a rope and for a column of air, and every time it does, whatever is waving travels at the v that equation contains.

Looking ahead

You now have both halves: what a travelling wave must look like, $f(x - vt)$, and why a medium carries one at a speed of its own choosing. **Note 8** sends two of them at each other. A wave travelling right meets its own reflection travelling left, and the sum of the two does something neither one does alone: it stops moving, and only certain wavelengths survive.

Where this sits. Section 2 is Note Note 8's derivation done to a piece of string instead of a block — the same Newton's second law, the same "write the net force and see what it is proportional to". The difference is that a string has neighbours, so the restoring force comes from the *curvature* rather than from a spring, and the result is an equation in two variables instead of one.

BACK TO YOUR PREDICTION

Tighter: faster. T_h is on top of $v^2 = T_h/\mu$. **Heavier string: slower.** μ is underneath. **Shaking faster: no change at all.** Neither the frequency nor the amplitude appears anywhere in the derivation — shake it faster and the wavelength shortens to keep λf equal to the speed the string already had. That is the same answer Note 1 got from a simulation and a ruler, and it is now a theorem rather than an observation.